Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	0157CY241012
Project Name	EV Market Analysis (Team: Data Warrior)
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Load the cleaned EV dataset (1,200 records, 2021-2026) from CSV into Tableau.	High
2	Data Processing	Clean data, fix data types, and create derived fields such as price per km and revenue.	High
3	Business Analysis	Calculate KPIs: total units sold, gross revenue, average range, customer rating and CO2 saved.	High
4	Business Insights & Analytical Outputs	Provide dashboards and charts: brand comparison, battery vs range scatter, range distribution and body-type split.	High
5	User Interaction & Analysis	Users can filter by brand, category, battery type and year, compare brands and drill down.	High
6	Data Update & Reporting	Dashboard is published on Tableau Public and the web app; data refresh by replacing the CSV and republishing.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Simple dashboard with clear filters; works on desktop and mobile browsers.
NFR-2	**Security**	Uses public, non-personal data only; no login or sensitive data is stored.
NFR-3	**Reliability**	Cleaned data with zero null or invalid values; results stay consistent.
NFR-4	**Performance**	Filters and charts respond in under 100 ms (average query 7.6 ms in testing).
NFR-5	**Availability**	Available online through Tableau Public and GitHub Pages.
NFR-6	**Scalability**	Handles about 100,000 records in the web app; larger data needs a data warehouse.